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Dedication

*To my dear parents,
whose unconditional love, patience, and sacrifices have lighted every step of my
journey. Nothing I achieve would have been possible without you.*

*To my family,
for their constant encouragement and unwavering support.*

*To my teachers and mentors,
who shared their knowledge with generosity and shaped the engineer I am
becoming.*

*To my friends and colleagues,
for the moments of doubt we overcame together and the joys we shared along
the way.*

To everyone who believed in me.

Mohammed El Badry

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Abstract

Mobile applications built by software consultancies such as **Theodo** must comply with the publication rules of the **Apple App Store and Google Play**. In practice, non-compliance is often discovered only at the very end of the development cycle, when the application is submitted to the stores. A late rejection then causes costly **rework and delivery delays** for applications already in production. This work, carried out as part of the **Mobile Software Factory** project, addresses this problem by shifting compliance verification to the left, that is, earlier and continuously throughout development.

The proposed solution is an **automated compliance audit tool**, delivered as a structured *skill* executed by an AI coding agent. Rather than a classic static analyzer, it relies on auditable policy packs that the agent applies through reasoning, following a five-step workflow : framework detection, review-oriented codebase understanding, business-domain selection, application of the relevant policies, and report generation. The approach is deliberately scope-aware, evaluating only the guidelines relevant to the features the application actually uses, and evidence-based, so that every finding is traced both to an observed code path and to a written policy rule.

Because the tool is driven by a language model, its output cannot be validated with classical unit tests. A behavioral evaluation suite was therefore built with **Promptfoo**, combining minimal fixture projects, paired compliant/non-compliant scenarios, **deterministic assertions, and rubric-based judgments** to guarantee that the audit identifies the right issues, cites the right evidence, and avoids hallucinated findings.

Keywords : Store compliance, Mobile applications, App Store, Google Play, Shift-left, AI agent, LLM, Skill, Permissions, Promptfoo, Proof of concept.

Résumé

Les applications mobiles développées par des cabinets de conseil tels que **Theodo** doivent respecter les règles de publication de l'**Apple App Store** et de **Google Play**. En pratique, la non-conformité n'est souvent détectée qu'à la toute fin du cycle de développement, au moment de la soumission de l'application sur les stores. Un rejet tardif entraîne alors, à grands frais, des **reprises et des retards de livraison** pour les applications déjà en production. Ce travail, réalisé dans le cadre du projet **Mobile Software Factory**, répond à cette problématique en déplaçant la vérification de conformité « vers la gauche », c'est-à-dire plus tôt et de manière continue tout au long du développement.

La solution proposée est un **outil d'audit de conformité automatisé**, livré sous la forme d'une *skill* structurée et exécutée par un agent de programmation basé sur l'intelligence artificielle. Plutôt qu'un analyseur statique classique, il s'appuie sur des paquets de règles auditable que l'agent applique par raisonnement, en suivant un processus en cinq étapes : détection du framework, compréhension du code orientée revue, sélection du domaine métier, application des règles pertinentes et génération du rapport. L'approche est volontairement ciblée, n'évaluant que les directives pertinentes pour les fonctionnalités réellement utilisées par l'application, et fondée sur des preuves, de sorte que chaque constat est rattaché à la fois à un comportement observé dans le code et à une règle écrite.

Comme l'outil est piloté par un modèle de langage, sa sortie ne peut être validée par des tests unitaires classiques. Une suite d'évaluation comportementale a donc été construite avec **Promptfoo**, combinant des projets de test minimaux, des scénarios appariés conformes / non conformes, **des assertions déterministes et des jugements basés sur des grilles d'évaluation** afin de garantir que l'audit identifie les bons problèmes, cite les bonnes preuves et évite les constats hallucinés.

Mots-clés : Conformité aux stores, Applications mobiles, App Store, Google Play, Shift-left, Agent IA, LLM, Skill, Permissions, Promptfoo, Preuve de concept.

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General Introduction

As the culmination of my pursuit of a state engineering diploma in Software Engineering, Big Data, and Cloud Computing, I undertook an end-of-study internship at **Theodo Morocco**, a digital consultancy firm recognized for delivering high-impact web and mobile software solutions for its clients. This internship offered the opportunity to confront a concrete industrial problem and to design, build, and evaluate a complete solution within a professional engineering environment.

A consultancy such as Theodo regularly builds mobile applications that must eventually be published on the Apple App Store and Google Play. Each store enforces a detailed set of review guidelines and developer policies, and any violation can lead to the rejection of an application. The recurring difficulty is that this verification happens too late : teams develop an application over several weeks or months, and the non-compliance is only discovered at submission time, at the worst possible moment, right before delivery. This *late detection* causes costly rework and delays, while applications already in production accumulate a growing *compliance debt* as store requirements evolve. The core insight behind this project is that compliance should be verified continuously, as the application is being built, rather than as a one-shot gate at the very end.

This work was carried out within the **Mobile Software Factory** project, a collaborative initiative at Theodo aimed at improving the reliability and the speed of mobile application delivery. The project brought together a cross-functional team composed of developers, a tech lead, and a project manager. My contribution focused on the store-compliance component, materialized as an automated audit tool that a developer can run as a command during development to obtain a clear and actionable compliance status of the application.

The solution is designed as a structured *skill* executed by an AI coding agent. It inspects the source code and configuration of a cross-platform mobile application, applies a set of auditable policy rules, and produces a report describing what was checked, what passes, what fails, and how each issue should be remediated. The approach is deliberately scope-aware, evaluating only the guidelines relevant to the features the application actually uses, and evidence-based, so that each finding is justified by an observed behavior and a written rule rather than by guesswork. Particular attention was given to the quality assurance strategy, since a tool driven by a language model cannot be validated through classical unit testing and required a dedicated behavioral evaluation methodology.

This report is structured into several chapters to provide a clear and progressive overview of my contributions :

Chapter I – Host Organization & Project Context introduces Theodo Morocco and its place within the international Theodo Group, the work culture that framed the internship, the project team, and the overall mission. It situates the internship in its professional environment before any technical detail is discussed.

Chapter II –
Chapter III –
Chapter IV –

CHAPTER I

Host Organization and Project Context

*Understanding the environment in which a solution is built
is the first step toward building the right solution.*

1.1 Introduction

This chapter presents the general context in which my end-of-study internship took place. It introduces Theodo Morocco as the host organization, its connection to the international Theodo Group, its work culture, and the values that guide its teams. It also presents the recruitment process, the integration experience, the project team, the overall mission of the internship, and an analysis of the existing compliance process that motivates this work.

The purpose of this chapter is not to describe the technical solution in detail, but to situate the internship in its professional environment. The internship subject belongs to the field of artificial intelligence applied to software quality, with a specific focus on automating the store-compliance auditing of mobile applications through an AI coding agent. Such an agent supports development teams by analyzing a codebase, reasoning over store policies, and reporting compliance issues early in the development cycle.

Working on such a subject requires more than technical experimentation. It requires a structured environment, clear communication, careful documentation, and continuous feedback. Theodo's culture provided a suitable framework for this type of mission, combining autonomy, rigor, collaboration, and a strong focus on practical impact.



FIGURE 1.1 – Theodo logo

1.2 Host Organization

Theodo Morocco is a technology consulting company specialized in building high-value digital solutions and supporting organizations in their transformation projects. The company was initially known as Nimbleways before joining the international Theodo Group in 2019. This integration connected the Moroccan team to a wider ecosystem composed of several specialized entities working in areas such as finance, healthcare, cloud, public services, data, and artificial intelligence.

Theodo Group follows a culture inspired by Lean Management. Its teams focus on continuous improvement, operational excellence, rapid feedback, and the delivery of measurable value. Work is organized around close collaboration with stakeholders, clear prioritization, and a constant effort to understand the real needs behind each project.

Within this ecosystem, Theodo Morocco plays an important role as a delivery and expertise hub. The company brings together profiles with strong analytical, product, and engineering skills, while maintaining a culture based on learning, ownership, and collective problem solving. This environment is particularly relevant for an internship related to artificial intelligence and agents, because such projects require both experimentation and disciplined execution.



FIGURE 1.2 – Theodo Group illustration

1.3 Values and Mission

Theodo's values play a central role in the way teams work on a daily basis. They are not abstract principles ; they are reflected in concrete practices such as task organization, feedback rituals, documentation, problem solving, and continuous improvement.

1.3.1 Rigor

Rigor means diligence and care in the work delivered. Clients expect the highest standards of professionalism, and this is reflected in concrete habits : daily to-do lists, regular reviews, systematic documentation of decisions, and continuous capitalization on learnings.

Mediocrity is not tolerated. This formality in processes helps ensure consistent and predictable results, sprint after sprint. In a compliance-auditing project driven by a language model, rigor is especially important because results must be reproducible, traceable to evidence, and documented in order to avoid approximate decisions or unsupported assumptions.

1.3.2 Pragmatism

Pragmatism consists of solving problems through structured methods rather than individual heroism. Discussions are grounded in concrete elements : data, observations, user feedback, and measurable results. Abstract debates have little value when a decision can be informed by an experiment, a prototype, or a real use case.

For a project based on an AI auditing agent, pragmatism helps keep the focus on usefulness. The objective is not simply to use an advanced technology, but to verify that the tool truly helps teams catch compliance issues earlier, reduce manual checking, and avoid costly store rejections.

1.3.3 Eagerness to Progress

Theodo considers problems as opportunities to learn. This eagerness to progress is notably expressed through the Andon system : any team member can signal a blocker in real time so that the group can address it collectively, without delay and without trying to hide the issue.

This culture encourages transparency and accelerates learning. It is highly relevant for a store-compliance topic, where store guidelines, models, and best practices evolve quickly. Continuous learning becomes necessary to keep the policy rules up to date, understand model limitations, and improve results through iteration.

1.3.4 Team Spirit

Team spirit means that the collective interest comes before individual interests. Team members challenge each other to maintain high standards, and project success is attributed to the group as a whole. This creates an environment where trust and high expectations coexist.

In a mission involving an AI auditing agent, team spirit is essential. It helps validate assumptions, challenge the policy rules and the review workflow, improve the clarity of the reports, and avoid designing a solution from a single isolated perspective.

These values have been recognized by the independent organization ChooseMyCompany, which distinguished Theodo for the quality of its work environment and employee satisfaction.



FIGURE 1.3 – Awards received by Theodo from ChooseMyCompany.

1.4 Recruitment Process

Obtaining the internship opportunity at *Theodo* involved a progressive recruitment process designed to evaluate technical reasoning, problem-solving ability, motivation, communication skills, and cultural fit.

The first step consisted of a technical assessment covering algorithmic thinking, logic, programming fundamentals, and problem solving. The objective was to evaluate the ability to reason clearly and structure a solution when facing different types of challenges.

The second step was an HR interview. This exchange allowed me to present my academic background, motivations, previous experiences, and interest in artificial intelligence topics. It was also an opportunity to better understand Theodo's culture, expectations, and working environment.

The process continued with more in-depth interviews focused on reasoning, communication, and the ability to explain a solution approach. These exchanges assessed not only knowledge, but also how I think, how I react to feedback, and how I approach unfamiliar problems.

The final discussions focused on the alignment between my profile, the company's values, and the expectations of the internship. This process confirmed my interest in joining an environment where quality, learning, collaboration, and user impact are central.

1.5 Integration and Onboarding Process

On my very first day at Theodo Morocco, I joined the **Asakai** meeting, the weekly ritual where each team presents its progress and the blockers it has encountered. Being placed in this ritual from the first morning immediately conveys the rhythm of the company.

The onboarding itself was organized around a starter project hosted on a **Kanban board in Notion**, Theodo's reference project-management tool. This project deliberately spans many areas of software development, from frontend to backend and testing. It is not meant to be easy : it is designed to confront newcomers with the company's standards as quickly as possible.

Several workshops complemented this immersion phase : an introduction to the **Andon** system, a presentation of the **Perfect User Story**, and a session on the **Daily Mail** as a tool for asynchronous communication.

What struck me most about this onboarding was its density. In a few weeks, you do not read how Theodo works, you live it.

1.6 Project Team

The project was carried out by a team of five people with complementary roles. Understanding who does what in this team is useful to grasp how the project was steered and on which contribution each person focused.



FIGURE 1.4 – The Mobile Software Factory project team

Member	Role	Main contribution
Hamza Berraho	Project Manager	Framed the scope of the project from the start and continued to arbitrate what the tool should, and should not, take charge of, keeping the mission focused and aligned with Theodo's delivery needs.
Yassine Ghazouan	Tech Lead	Guided the team throughout the project, validated the architectural decisions, and challenged the design and implementation choices at every stage, directly shaping the technical trade-offs behind the tool.
Mohammed Bourhym	Software Engineer	Contributed to the design and architecture of the auditing solution and to its development, in close collaboration with the rest of the engineering team on the technical topics.
Ismail Drissi	Software Engineer	Contributed to the design and development of the solution, including the policy references and the testing effort that supports the reliability of the tool.
Mohammed El Badry	Software Engineer (author)	Worked on the design, architecture, development, and behavioral evaluation of the compliance solution from the first day until delivery, collaborating continuously with the Tech Lead and the Project Manager to adjust the implementation choices and steer the tool in the right direction.

TABLE 1.1 – Members of the project team and their contributions

1.7 Internship Mission

The mission entrusted to me during this internship is the design and development of the store-compliance component of the **Mobile Software Factory**, an internal tool that audits mobile applications against the publication rules of the Apple App Store and Google Play. The tool is run by a developer as a command during development : it inspects the source code and configuration of a cross-platform mobile application, applies the relevant store policies, and produces a clear report stating what was checked, what passes, what fails, and how each issue should be fixed.

Two design principles are central to the mission. The tool is **scope-aware** : it only evaluates the guidelines relevant to the features the application actually uses, so teams are not drowned in irrelevant rules. It is also **evidence-based** : every finding must be traced both to an observed code path and to a specific written policy rule, and when the evidence is insufficient, the finding is marked as unclear rather than guessed.

The most structuring objective of the project is to **shift compliance left**, that is, to detect issues early and continuously during development rather than at submission time. As a proof of concept, the scope was deliberately focused on the **permissions** domain, for both Android

and iOS, on the Flutter, React Native, and Expo frameworks, with an architecture designed to welcome additional compliance domains later.

1.8 Analysis of the Existing Compliance Process

1.8.1 A critical but late-handled step

Store compliance is not a peripheral concern in mobile development. An application that violates a single store guideline can be rejected at submission, regardless of the quality of the rest of the work. Yet, in practice, this verification happens at the very end of the cycle : teams develop an application over weeks or months, and compliance is only checked when the application is uploaded to the stores. A rejection discovered at that point arrives at the worst possible moment, right before delivery.

1.8.2 A manual, variable, and error-prone check

The problem is not that compliance checks exist ; it is the way they are performed. The Apple App Store Review Guidelines and the Google Play policies are long, detailed, and constantly evolving. Verifying them manually is repetitive and leaves room for omissions, especially when several applications are being developed in parallel. The result also depends heavily on the person performing the review : from one developer or one project to another, the depth and the consistency of the check vary significantly, and no automatic mechanism guarantees that every relevant rule has been considered.

1.8.3 Impact on the delivery rhythm

This late and manual approach has a direct impact on the delivery rhythm. A late rejection forces the team to revisit code and configuration that were considered finished, causing rework and delivery delays. Applications already in production accumulate a growing compliance debt as store requirements evolve, and risk removal if they fall out of conformity. Compliance therefore becomes a recurring, hidden bottleneck whose cause is not immediately visible.

1.9 Conclusion

This chapter set the organizational and project context of my internship at Theodo Morocco. The company, with its Lean culture, its agile practices, and its demand for software quality, provides an environment in which a project such as the Mobile Software Factory fits naturally.

The four values of Theodo, namely rigor, pragmatism, eagerness to progress, and team spirit, form a solid foundation for a mission that requires both experimentation and disciplined, evidence-based execution.

The analysis of the existing compliance process confirms the reality of the problem : a manual, late, and variable verification that weighs on the delivery rhythm and exposes teams to costly store rejections. The following chapter presents the methodology adopted and the way the proposed solution fits into the development workflow.

CHAPTER II

Methodology & Development Approach

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